# Chase Tremain

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## EDUCATION

**University of Michigan - College of Engineering** **Ann Arbor, MI**
*Master of Science in Mechanical Engineering* *Class of 2027*
Prospective Courses: Microelectromechanical Systems, Vehicle Control Systems, Fundamentals of Vehicle Dynamics

**University of Michigan - College of Engineering** **Ann Arbor, MI**
*Bachelor of Science in Mechanical Engineering* *Class of 2026 - G.P.A 3.6*

## WORK EXPERIENCE

**University of Michigan Electric Vehicle Center** **Ann Arbor, MI**
*Research Assistant – Project Lead* *August 2025 - Present*
- Designed and built a hardware-in-the-loop EV testbench capable of simulating vehicle thermal loads to determine compressor efficiencies for industry sponsors.
- Developed and validated MATLAB/Simulink models to analyze refrigerant cycle behavior, heat exchanger loads, mass flow rates, tank temperature balances, compressor efficiency, and PID pump control.
- Managed project planning, component sourcing, and system specifications while maintaining compliance with safety and performance requirements within a $50,000 project budget.

**SPX Technologies - Cincinnati Fan** **Mason, OH**
*Manufacturing and Quality Intern* *May 2025 - August 2025*
- Programmed and optimized YASKAWA ArcWorld 2000 robotic welding arm, reducing cycle times by 8%, and programmed the first robotically welded wheel to cut direct labor by 80%.
- Designed and implemented custom welding and testing fixtures using machine design formulas and CAD principles.
- Performed manufacturing process development by analyzing wire feed speed, travel speed, and heat input against part geometry to establish optimal programming parameters.

**Spark Electric Racing** **Ann Arbor, MI**
*Mechanical Engineer* *September 2022 – May 2026*
- Conducted research on diverse methods for mounting various subsystems to an electric motorcycle, including identifying necessary parts and a variety of sources for obtaining the parts.
- Used advanced CAE software to analyze how a motorcycle chassis reacts to real-world conditions.
- Worked on a team to develop, design, and build a weight-efficient chassis for an electric racing motorcycle.

**Kamps Pallets Inc.** **Grand Rapids, MI**
*Continuous Improvement Intern* *May 2024 – April 2025*
- Applied lean manufacturing principles to conduct material utilization analysis to identify and eliminate unnecessary waste, resulting in a 10% productivity improvement.
- Worked in a small group to develop and implement software to assist managers companywide in completing essential tasks and surpassing production goals.
- Performed inventory management surveys to optimize stock levels and redistribute unused material to other locations nationwide.
- Conducted time studies and workflow analysis to identify optimal production rates and fabrication target goals for dozens of employees working across thirteen different machines.

## SKILLS

- *MATLAB/Simulink, GT Suite, Microsoft Office Suite, SolidWorks, CATIA, Inventor, Hyperworks, Ansys*